

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month in Earth in explore deep biospheres, in Space Age we learn about the Twinkle Mission, in R&D we look at some next-gen wearables being developed, and more.



Star turns into disk

Researchers have observed a black hole in the process of swallowing a star, and seen the formation of an accretion disk, where the material from the star gets deformed into a rotating disk around the black hole.

<http://dgit.in/accdisc>

WHAT'S NEW

Monster crocodile ate dinosaurs

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0 million years ago, unsuspecting dinosaurs heading to the banks of a river to drink water would have been chewed up by this ancient crocodilian, known as the Deinosuchus. The monster, a relative of the modern day alligator, is one of the largest, if not the largest, known crocodilian. Between 82 to 75 million years ago, they inhabited coastal wetlands, and were the largest predators in the ecosystem, growing to 10 meters in length, and outweighing the other carnivorous dinosaurs. It had a wide gaping jaw, like modern crocodiles, with teeth that could reach the size of bananas. Researchers already suspected that this thing ate dinosaurs based on teeth marks in the fossil remains of



dinosaurs, as well as the cranial remains of the creature itself. Now researchers have shown that the size of the creature and the strength of its jaws

were enough to eat dinosaurs. They identified bite marks on fossil dinosaur bones and turtle shells. The creature has a large and prominent nose, which is unlike extant alligators or crocodiles, and scientists are baffled about why the creature's snout is shaped in such a way. The creature also had two holes of unknown purpose on the snout.

<http://dgit.in/dinoeater>

New telescope to hunt for rogue exoplanets

Planet hunting telescopes have identified so many exoplanets that we now know that there is at least one planet in orbit around every star. When planets are being formed in the accretion disks around young stars, the gravitational interactions between these bodies could eject some planets straight out of the system, and into interstellar space. This could lead to a large population of rogue

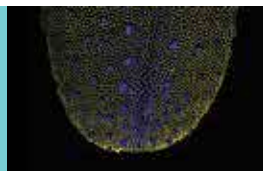
planets, and till now, there was no technology for actually detecting them. Most of the exoplanets we know of have



been detected through what is called the transit method, where the planet has to pass in front of the host star in rela-

tion to the Earth. Now, a new telescope being launched by NASA, called the Nancy Grace Roman Space Telescope, which will be the first to conduct a dedicated survey of these rogue planets, and we may finally find out if the number of rogue planets in the galaxy matches the predictions. Apart from rogue exoplanets, the telescopes will hunt for regular exoplanets as well.

<http://dgit.in/grt>



Test mice taste nice

According to conventional wisdom, the taste cells on tongues can only detect one or two flavours at max, such as sweet, salty or sour, and are very specific. Now researchers have come across multitasking taste cells on mice that can sense upto four different stimuli at once.

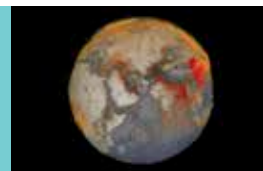
<http://dgit.in/micetaste>



Cuttlefish hides its colours

The cuttlefish is a sea creature that is popular on the internet, as well as aquariums because of how colourful and flamboyant it looks. However, scientists are now setting the record straight, and noting that in the wild, it is almost always camouflaged, and nearly impossible to find.

<http://dgit.in/fabfish>



Methane emissions soar

Global emissions of methane have reached an all time high, which has been the trend for the past few years. Methane, a greenhouse gas that is more effective at trapping heat than carbon dioxide, is released into the atmosphere from coal mining, oil production, and agriculture.

<http://dgit.in/mggw>